A Complete and Natural Rule Set for Multi-Qudit Clifford Circuits in All Odd Prime Dimensions

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Outline

This talk

- A presentation by generators and relations of the n-qudit Clifford group, for every odd prime dimension p: 16 relations, none involving more than 3 qudits.
- A unique normal form for symplectic (= Clifford modulo Pauli) circuits.
- All, except for two math facts, are formalised in Agda.

About this slides

- Every Agda code block is typechecked by Agda.
- Written by *Claude* and Xiaoning Bian ($\sim$200-line outline prompt and slides checking $\&$ improvement).

Introduction

From qubit to qudit Clifford gates

Fix an odd prime p and let $\omega = e^{2\pi i/p}$. The Clifford group $\mathcal{C}_n$ is generated by:

	qubit ($p = 2$)	qudit (odd prime p)
H	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$\lvert j\rangle \mapsto \frac{1}{\sqrt{p}} \sum_k \omega^{jk} \lvert k\rangle$
S	$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	$\lvert j\rangle \mapsto \omega^{j(j-1)/2} \lvert j\rangle$
CZ	$\mathbf{diag}(1, 1, 1, -1)$	$\lvert j\rangle\lvert l\rangle \mapsto \omega^{jl} \lvert j\rangle\lvert l\rangle$

Note the $/2$ in the qudit S is the inverse of 2 in $\mathbb{Z}_p$. A global phase can be added to the definition of H to facilitate computation.

The setting: one parameterized Agda module

These slides are a single Agda module, parameterised by a prime p and a multiplicative generator g of $\mathbb{Z}_p^*$:

```
module qpl26slides.talk
  (p-3 : ℕ)
  (let p-2 = 1+ p-3)
  (p-prime : Prime (3+ p-3))
  (let open PrimeModulus' p-2 p-prime)
  (g*@(g , g≠0) : ℤ* p)
  (g-gen : ∀ ((x , _) : ℤ* p) → ∃ \ (k : ℤ p-1) → x ≡ g ^′ toℕ k)
  where
```

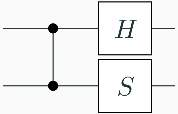
- $p =$ `p-3` $+ 3$: every odd prime, uniformly.
- Everything downstream — rule sets, normal forms, theorems — is parameterised by the choice of g.

Circuits in Agda

A gate set is a family of gate types indexed by arity; a circuit on n wires is a word over wire-indexed generators.

```
data SympGate : ℕ → Set where
  H-gate  : SympGate 1
  S-gate  : SympGate 1
  CZ-gate : SympGate 2

example : Circuit 2
example = CZ • H ↑ • S
```

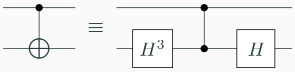
- `Circuit` n = `Word` (`Gen` n): free monoid; `_↑` shifts a circuit up one wire.
- diagrams read in matrix-multiplication order: `example` =

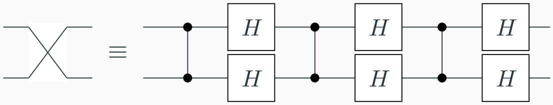


Derived gates: the rest of the usual gate set

X, Z, the swap Ex, and the two controlled-X gates are *words*, not extra generators:

```
Z X : ∀ {n} → Word (Gen (₁₊ n))
Z = H • H • S • H • H • S⁻¹
X = H • S • H • H • S⁻¹ • H

Ex CX XC : ∀ {n} → Word (Gen (₂₊ n))
Ex = CZ • H ↓ • H ↑ • CZ • H ↓ • H ↑ • CZ • H ↓ • H ↑
CX = H ↓ ^ 3 • CZ • H ↓
XC = H ↑ ^ 3 • CZ • H ↑
```





Main results

Main result 1: a presentation of the qudit Clifford group

The following 16 realtions presents the projective Clifford group $\mathcal{C}_n/\langle\omega\rangle \;\cong\; \mathrm{Pauli}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p)$:

```
Theorem-1 : ∀ (n : ℕ) → (n CRel,_===_) IsPresentationOf (Pauli⋊Sp n)
Theorem-1 = PapPres.presentation
```

- `_CRel,_===_` is the 16-relation rule set, shown on the next slides.
- The theorem holds *at every width* n, for *every* odd prime p and every multiplicative generator g — the whole deck is parameterised by them.

The 16 relations, in Agda (one qudit)

```
data _CRel,_===_ : (n : ℕ) → WRel (Gen n) where
  order-S     :       (₁₊ n) CRel, S ^ p === ε
  order-H     :       (₁₊ n) CRel, H ^ 2 === M₋₁
  M-power     : ∀ k → (₁₊ n) CRel, Mg^ k === M (g^ k)
  semi-MR     :       (₁₊ n) CRel, R • Mg === Mg • R^ (g * g)
  order-SH    :       (₁₊ n) CRel, (S • H) ^ 3 === ε
  comm-HHSHHS :       (₁₊ n) CRel, H • H • S • H • H • S === S • H • H • S • H • H
```

- M_x is the multiplier $|j\rangle \mapsto |xj\rangle$, spelled as the word $R^{x^{-1}}\, H\, R^x\, H\, R^{x^{-1}}\, H$ with $R = S \cdot Z^{1/2}$.
- One-qudit relations talk about the powers of g — this is where odd primes differ from the qubit case.

The 16 relations, in Agda (two qudits)

```
order-CZ   : (2+ n) CRel, CZ ^ p === ε
order-Ex   : (2+ n) CRel, Ex ^ 2 === ε
comm-CZ-S↑ : (2+ n) CRel, CZ • S ↑ === S ↑ • CZ
semi-M↑CZ  : (2+ n) CRel, CZ • Mg ↑ === Mg ↑ • CZ^ g
semi-Ex-S↑ : (2+ n) CRel, Ex • S ↑ === S ↓ • Ex
semi-Ex-H↑ : (2+ n) CRel, Ex • H ↑ === H ↓ • Ex
blake-c12  : (2+ n) CRel, (S ^ p-1) ↑ • (S ^ p-1) ↓ • CX ^ p-1 • S ↓ • CX === CZ
```

- `blake-c12` relates two circuits that implement the Controlled-Z operator.

The 16 relations, in Agda (three qudits)

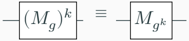
```
yang-baxter : (₃₊ n) CRel, Ex ↑ • Ex ↓ • Ex ↑ === Ex ↓ • Ex ↑ • Ex ↓
cz-slide    : (₃₊ n) CRel, Ex ↓ • Ex ↑ • CZ === CZ ↑ • Ex ↓ • Ex ↑
semi-CX↑-CZ↓ : (₃₊ n) CRel, CZ ↓ • CX ↑ === CZ02 • CX ↑ • CZ ↓
```

- Only these 16 are axioms. $XZ = ZX \pmod{\omega}$, $\mathbf{\mathit{Ex}}$-vs-CZ, and both Pauli-vs-CZ rules are *derived*.
- Plus, for every gate set, the same three monoidal structural rules: congruence under $\uparrow$, gates on disjoint wires commute, and scalars commutes with every gate.

The 16 relations, as circuits (1–2 qudits)

order-S


order-H


M-power


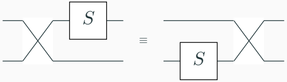
semi-MR

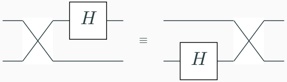

order-CZ


order-Ex

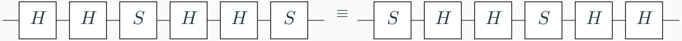

comm-CZ-S↑


semi-M↑CZ


semi-Ex-S↑


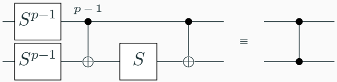
semi-Ex-H↑


order-SH

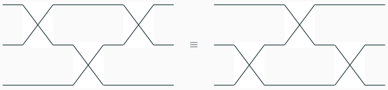

comm-HHSHHS


The 16 relations, as circuits (2–3 qudits)

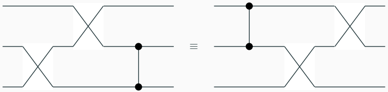
blake-c12



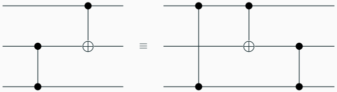
yang-baxter



cz-slide



semi-CX↑-CZ↓



What "presents" means

```
record _Presents_ (_===_ : WRel X) (G : Group a ℓ) : Set (a ⊔ ℓ) where
  field gl : Grouplike _===_
  module GL = Group-Lemmas _===_ gl
  open GroupMorphisms (Group.rawGroup GL.•-ε-group) (Group.rawGroup G)
  field
    ⟦_⟧ : Word X → Group.Carrier G
    iso : IsGroupIsomorphism ⟦_⟧
```

$$\mathsf{Cir}\ n \,/ \approx_{\text{16 rules}} \xrightarrow[\cong]{[\![\_]\!]} \mathrm{Pauli}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p) \;\cong\; \mathcal{C}_n / \langle \omega \rangle$$

- *Soundness*: $[\![\_]\!]$ is well defined — each rule holds semantically.
- *Completeness*: $[\![\_]\!]$ is injective — semantically equal circuits are related by the rules.

Main result 2: a unique normal form for symplectic operators

Symplectic operators are Clifford mod Pauli: the “proper” part. Normal forms are data, built from four atomic boxes:

```
A B D E : Set

A = Σ[ ab ∈ (ℤ p × ℤ p) ]
  (ab ≢ (0 , 0))

B = ℤ p × ℤ p

D = ℤ p × ℤ p

E = ℤ p
```

$A_{a,b}$, $a \neq 0$

$A_{0,b}$, $b \neq 0$

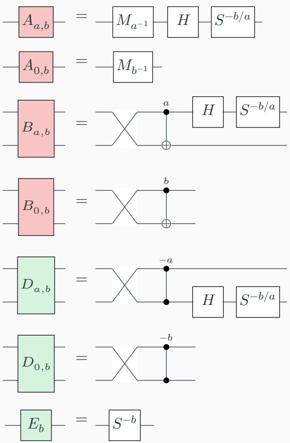
$B_{a,b}$, $a \neq 0$

$B_{0,b}$

$D_{a,b}$, $a \neq 0$

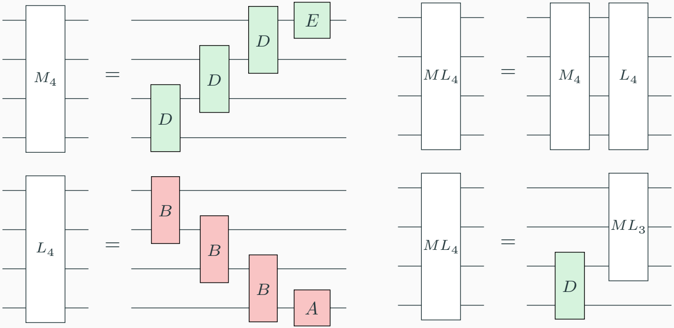
$D_{0,b}$

E_b



Rows, and the staircase

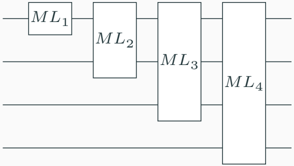
```
M L ML : ℕ → Set

M 0 = ⊤

M (1+ n) = Vec D n × E

L 0 = ⊤

L (1+ n) = Vec B n × A

ML 0 = ⊤

ML 1 = M 1 × L 1

ML m@(2+ n) =

  M m × L m ⊎ D × ML (1+ n)
```



The normal form, as a circuit

Reading normal forms back as circuits is structural recursion:

```
NF : ℕ → Set
NF 0 = ⊤
NF (1+ n) = NF n × ML (1+ n)

[_]ⁿᶠ : ∀ {n} → NF n → Word (Gen n)
[_]ⁿᶠ {0}    tt        = ε
[_]ⁿᶠ {1+ n} (nf , ml) = [ nf ]ⁿᶠ ↑ • [ ml ]ᵐˡ
```



Everything is Agda-checked

- ≈ 330 files, $\approx 74\,000$ lines of Agda, all `--safe`, no postulates.
- Claude did the semantic side code and the refactorization and improvment of the syntactic side.

What the formalisation covers

Presentations, normal forms, rewriting to normal forms, completeness.

What it does not

The two glue facts that the projective Clifford group is $\mathbf{Pauli} \rtimes \mathbf{Sp}$, and that the Clifford group is a central extension of it by $\langle\omega\rangle$, which are well-known.

Method

Divide and conquer

Factor the group; present the factors; compose the presentations.

$$1 \to \mathbf{Pauli}_n \to \mathcal{C}_n/\langle\omega\rangle \to \mathbf{Sp}(2n, \mathbb{Z}_p) \to 1$$

semidirect: it splits

$$1 \to \langle\omega\rangle \to \mathcal{C}_n \to \mathcal{C}_n/\langle\omega\rangle \to 1$$

central extension

- Doesn't this just move the difficulty around? No: the symplectic factor is where the real rewriting happens, and keeping the Pauli part out of it pays off everywhere.
- Prior work (qubit: Selinger; qutrit: Li et al.) worked with the whole Clifford group at once — the derivations drag a "messy" Pauli correction through every step.

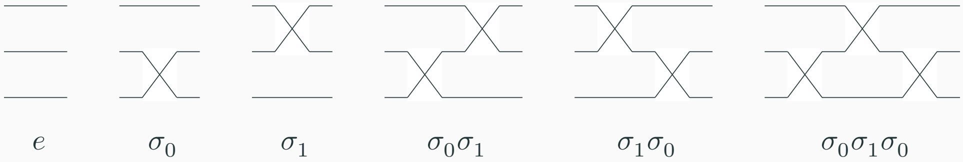
Symplectic presentation: the plan

Finding a complete rule set for $\mathbf{Sp}(2n, \mathbb{Z}_p)$, in seven steps:

1. Normal form — boxes, rows, and the staircase;
2. Rewrite to the normal form — the coset updating table, warming up with S_n;
3. From symmetric to symplectic — the same framework, under a richer action;
4. Uniqueness of the normal form;
5. Relation reduction — the 67 relations the rewrite uses, derived from 18;
6. The symplectic presentation — the theorem;
7. Comparison with the qubit case.

Warm-up: the symmetric group S_n

Swap circuits on n wires present S_n. All of S_3:



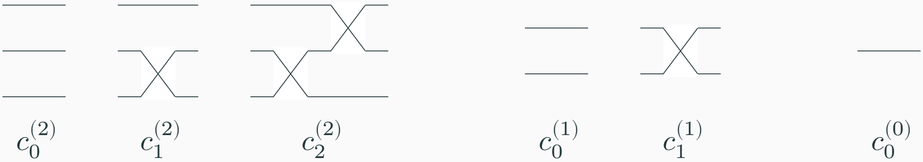
Circuits have a built-in tower of subgroups

$$\mathsf{Cir}\,0 \;<\; \mathsf{Cir}\,1 \;<\; \cdots \;<\; \mathsf{Cir}\,n$$

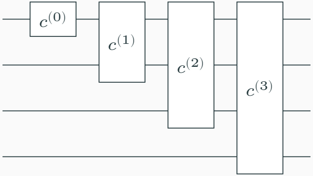
($\mathsf{Cir}\,k$ = circuits that only touch the top k wires). For S_n: $S_0 < S_1 < \cdots < S_n$.

Coset representatives, and the normal form

Representatives C_2 of S_3/S_2 (and of S_2/S_1, S_1/S_0): the staircases



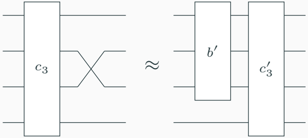
- For $f \in S_3$ there is a unique $c \in C_2$ with $c \circ f^{-1}$ fixing the bottom wire, i.e., the c that sends $f^{-1}(0)$ to 0.
- Recurse on $c \circ f^{-1} \in S_2$: every f is uniquely $c^{(0)} \cdot c^{(1)} \cdot c^{(2)}$.
- In general, given $G_0 < \cdots < G_n$ and coset reps C_k of G_k/G_{k-1}, elements of G_n factor uniquely as $\prod_k c_k$ — a mixed-radix encoding: $|S_3| = 3 \cdot 2 \cdot 1$.



Rewriting to the normal form: the coset updating table

```
ract : C (1+ n) → Gen (2+ n) → Circuit (1+ n) × C (1+ n)
ract {n} ε        σ-gen       = ε , σ• ε
ract {n} (σ• ε) σ-gen       = ε , ε
ract {1+ n} (σ• σ• c) σ-gen = (σ {n = n}) , σ• σ• c
ract {n} ε        (g ↥)       = [ g ]ʷ , ε
```

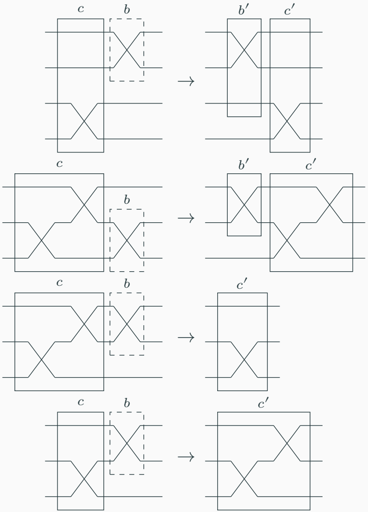
```
ract-sound : ∀ {n} c b →
  let open PB ((2+ n) VRel,_===_)
      (b' , c') = ract {n} c b
  in  [ c ]ᶜ • [ b ]ʷ ≈ b' ↑ • [ c' ]ᶜ
```



Pushing a generator b through $c_3 \in C_3$: a new coset c_3', and a residual "dirty" circuit b' on the wires above. Then we recurse on pushing b' through C_2.

Rewriting to the normal form: push a generator through

Multiply a normal form by a generator; push it through the staircase:



disjoint wires: commute past

Yang–Baxter: pass through, shift up

$\sigma^2 = e$: staircase shrinks

staircase grows

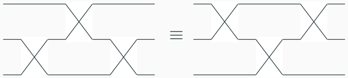
A presentation of S_n

Collecting the push-through relations and simplifying leaves just two rules (plus the structural ones):

```
data _SRel,_===_ : (n : ℕ) → WRel (Gen n) where
  order       : (2+ n) SRel, σ • σ === ε
  yang-baxter : (3+ n) SRel, σ • σ ↑ • σ === σ ↑ • σ • σ ↑
```

order

yang-baxter



```
Theorem-Sn : ∀ {n} → (n VRel,_===_) IsPresentationOf (Permutation′-group n)
Theorem-Sn = SymP.presentation
```

From S_n to $\mathrm{Sp}(2n, \mathbb{Z}_p)$: change the action

The same framework, three ways of looking at the wires:

1. S_n acts on $\{0, \ldots, n-1\}$: a wire is a *position*; $c \circ f^{-1}$ fixes 0 represented by the bottom wire.
2. S_n acts on **Vec Bit** n: a wire is a *bit*; $c \circ f^{-1}$ fixes the $\mathbb{Z}_2$-linear subspace represented by the bottom wire.
3. Adding H, S, CZ: the group acts on **Vec Pauli**$_1$ n — a $2n$-dimensional symplectic $\mathbb{Z}_p$-vector space; a wire is $\mathbf{Pauli}_1$, a 2-dimensional symplectic $\mathbb{Z}_p$-vector space.
4. Enlarging the coset layer from staircases to $\boldsymbol{ML}$ boxes, for each $f \in \mathbf{Sp}(2n, \mathbb{Z}_p)$, there exists $c \circ f^{-1}$ fixes the symplectic subspace of the bottom wire.

The framework applies to *any* $\mathbb{N}$-indexed family of groups with such a layered structure (`Circuit.CosetNF`, `Circuit.Uniqueness`).

Uniqueness of normal form

```
Theorem-2 : ∀ n {u v : NF n} → ⟦ [ u ] ⟧ ≈ˢ ⟦ [ v ] ⟧ → u ≡ v
Theorem-2 = U.⟦[]⟧-injective
```

$$
\begin{tikzcd}
\text{Cir}\, n \,/ \approx \arrow[rr, "[\![\_]\!]"] \arrow[d, bend left, "nf"] & & \mathrm{Sp}(2n, \mathbb{Z}_p) \\
NF\ n \arrow[u, bend left, "{[\_]}"] \arrow[urr, dashed, "\text{injective}"'] & &
\end{tikzcd}
$$

- Soundness + unique normal forms ⇒ completeness (`by-normalization`).
- The hard and interesting part is to find such *coset representatives* parameterized by the number of wires.

From 67 symplectic relations to 18

1. Rewrite *every* circuit to normal form; collect every relation the rewrite ever uses: 67 relations (box-vs-gate commutations, case by case).
2. By inspection, find a small set of primitive, bidirectional relations that derive all 67.
3. The choice is not canonical — pick the set you find natural.

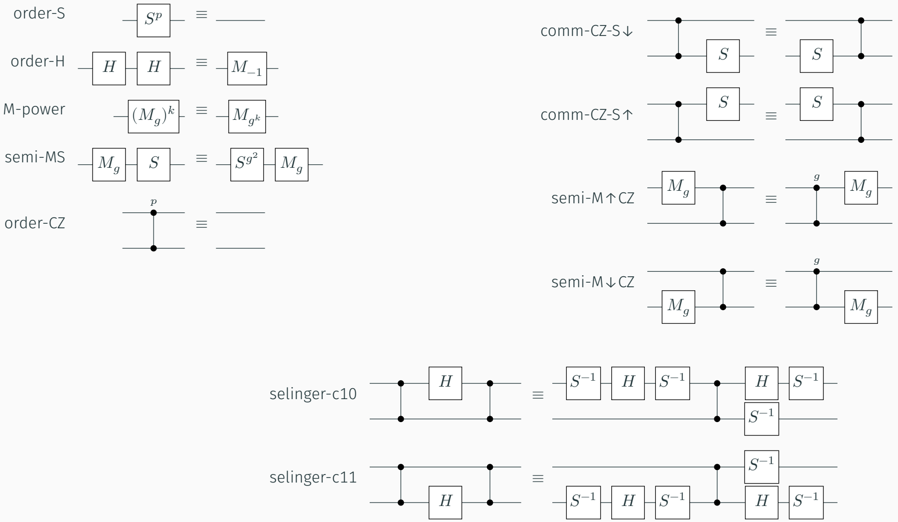
An example of a derived rule, $M_x \cdot M_y = M_{xy}$:

$$-\boxed{M_x}-\boxed{M_y}- \approx -\boxed{M_{xy}}-$$

```
M-mul : ∀ n (x y : ℤ* ₚ) → let open PB (Sim._QRel,_===_ (₁₊ n)) in
        M x • M y ≈ M (x *' y)
M-mul n = SimL.Lemmas1.lemma-M-mul n
```

($x = g^k$, $y = g^l$; `M-power` twice; add exponents mod p.)

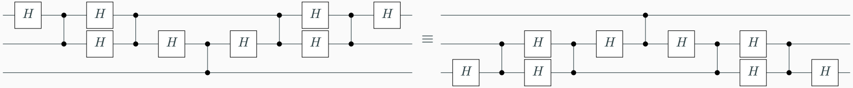
The simplified symplectic rule set (1–2 qudits)



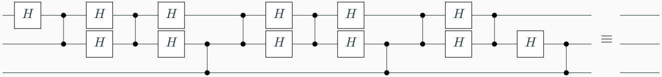
The simplified symplectic rule set (3 qudits), and the theorem

selinger-c12

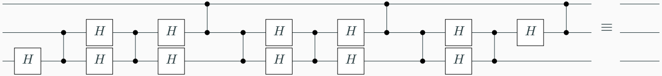
selinger-c13



selinger-c14



selinger-c15

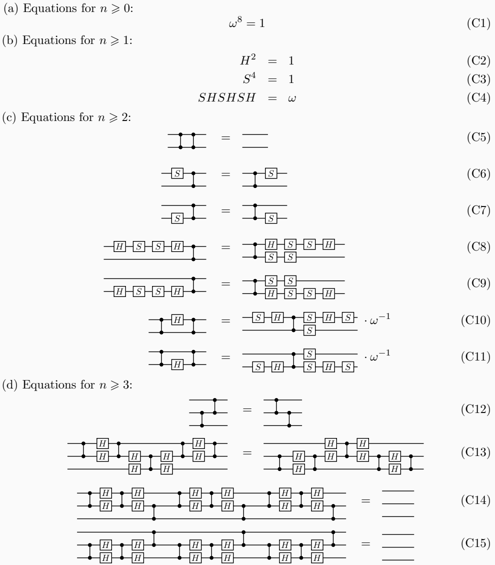



```
Theorem-Sp : ∀ {n} → (n SRel,_===_) IsPresentationOf (Sp-group n)

Theorem-Sp = SimPres.presentation
```

Comparison with the qubit rule set (Selinger)

- identical 3-qudit relations (C12–C15);
- similar 2-qudit relations;
- more complicated 1-qudit relations — the odd-prime multiplier calculus of M_g has no qubit counterpart.

Figure 8 of *Generators and relations for n-qubit Clifford operators* →



Composing: the plan

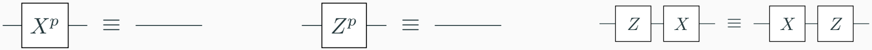
From the symplectic presentation to the Clifford presentation, in four steps:

1. The Pauli factor — presenting $(\mathbb{Z}_p \times \mathbb{Z}_p)^n$;
2. The semidirect product — gluing the Pauli and symplectic presentations along the conjugation action;
3. Plumbing — simplifying to the 16-relation presentation over H, S, CZ alone;
4. Getting the scalars back — the central extension by $\langle\omega\rangle$.

Composing: the Pauli factor

The projective Pauli group $(\mathbb{Z}_p \times \mathbb{Z}_p)^n$ has an easy presentation over generators X, Z on each wire:

```
data _PRel,_===_ : (n : ℕ) → WRel (Gen n) where
  order-X  : (₁₊ n) PRel, X ^ p === ε
  order-Z  : (₁₊ n) PRel, Z ^ p === ε
  comm-Z-X : (₁₊ n) PRel, Z • X === X • Z

Theorem-Pauli : ∀ {n} → (XZ._QRel,_===_ n) IsPresentationOf (+ₚ-group n)
Theorem-Pauli = XZ.presentation
```



Composing: the semidirect product

Semidirect product presentation: relations of both factors, plus twisted commuting: $h\,n'\,h^{-1} = \mathsf{conj}\,h\,n'$, so $(n,h)\cdot(n',h') = (n\,(\mathsf{conj}\,h\,n')\,,\;h\,h')$:

```
conj : Sym.Gen n → XZ.Gen n → Word (XZ.Gen n)
conj Sym.H-gen  X-gen     = Z
conj Sym.H-gen  Z-gen     = X ^ p-1
conj Sym.S-gen  X-gen     = X • Z
conj Sym.S-gen  Z-gen     = Z
conj Sym.CZ-gen X-gen     = X • Z ↑
conj Sym.CZ-gen (X-gen ↥) = X ↑ • Z

Pauli⋊Symplectic : (n : ℕ) → WRel (XZ.Gen n ⊎ Sym.Gen n)
Pauli⋊Symplectic n = (n PRel,_===_) ⋄ (n SpRel,_===_) ⋄ ConjRelʷ conj
```

Plumbing: presentation simplification

Two presentations are equivalent when the presented groups are isomorphic — each side's axioms derived from the other's. E.g. `Paper-V0.Iso`, `Simplified.Iso`.

	`SemiDirect`	`Paper-V0`
generators	X, Z, H, S, CZ	H, S, CZ
relations	Pauli + symplectic + conjugation	16 relations
generator reduction: X, Z become the derived words		
relation reduction: Pauli $+$ conjugation rules become derivable		

E.g., $CZ \cdot X\uparrow \approx X\uparrow \cdot Z\downarrow \cdot CZ$ becomes derivable.

Getting the scalars back

The Clifford group is a central extension of the projective Clifford group by scalars $\langle \omega \rangle$. By modifying the projective Clifford presentation, we get the Clifford presentation:

1. add a generator ω;
2. add a relation $\omega^p = \varepsilon$;
3. add relations that ω commutes with every gate;
4. correct each projective Clifford relation by the scalar it actually picks up — only `order-SH` needs one:

$$-\boxed{S}-\boxed{H}-\boxed{S}-\boxed{H}-\boxed{S}-\boxed{H}- \quad \equiv \quad \omega^{-1/8}$$

(where $1/8$ is the inverse of 8 in $\mathbb{Z}_p$)

Conclusion

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- Main result 1: presentations by generators and relations of the qudit Clifford operators and qudit Symplectic operators: all relations involve at most 3 qudits.
- Main result 2: a unique normal form for qudit Clifford and Symplectic operators.
- All are checked in Agda — except for the two glue facts: projective Clifford is a semidirect product and Clifford is a central extension of it by scalars.

Thank you!